

Scientific Programming Simplified with GNU Scientific Library

GSL, the GNU Scientific Library, is written entirely in C, and contains a set of numerical computing routines. Scientific programmers who appreciate the power of C and use it instead of higher-level languages will find it easy to adopt GSL. It has routines covering a lot of operations like complex numbers, roots of polynomials, vectors and matrices, permutations and combinations, interpolation and much more. This article guides you through installing GSL and compiling your first program with it.

Most people who rely heavily on computers to solve complex mathematical problems seem to be extremely frustrated while writing their programs. I believe, and you will no doubt agree, that the biggest hurdle in programming is detecting errors. Most of the time, we may not see syntax errors or any recognisable mistake—but the answer may still be wrong.

Recently, my friend Sanket asked me if there was an easier way around this problem. I replied, “Of course! There is GSL to rescue

you!” He seemed to be hearing of GSL for the first time; I actually found it surprising that such an amazing library is not yet quite popular among scientists and engineers.

The GNU Scientific Library will no doubt appeal to those who appreciate the power of C, and prefer it to higher-level languages. Being Free Software, everyone is free to use GSL, and to redistribute programs built on it. The basic question is, “How will GSL make my work easier?” GSL is meant for ordinary scientific programmers who know C at a fair level of complexity. GSL routines cover a lot